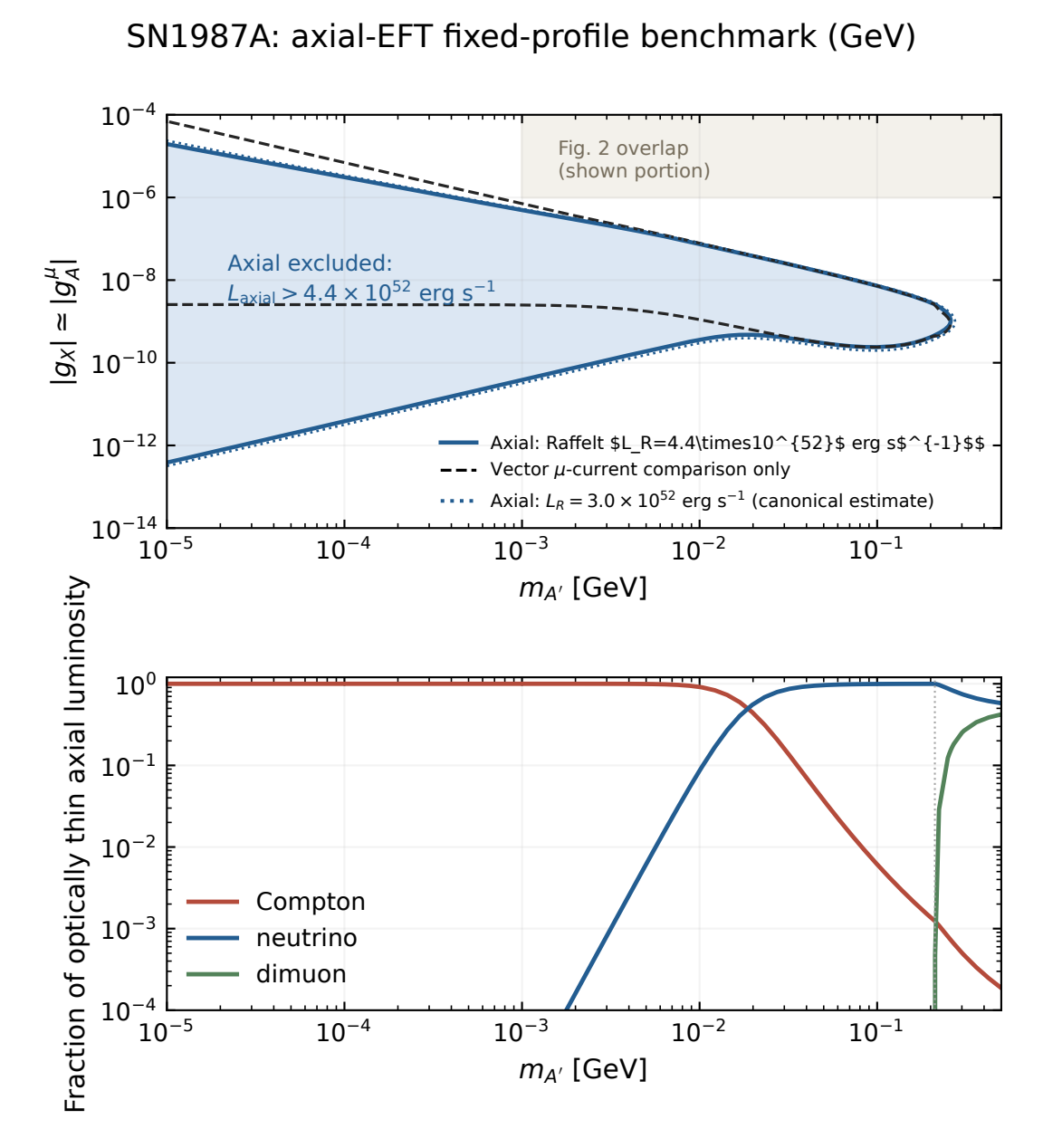
SN1987A: axial-EFT fixed-profile benchmark (GeV)
Fig. 2 overlap
(shown portion)
Axial excluded:
$L_{\rm axial} > 4.4 \times 10^{52}$ erg s^{-1}
Axial: Raffelt $L_R=4.4\times10^{52}$ erg s^{-1}
Vector μ-current comparison only
Axial: $L_R = 3.0 \times 10^{52}$ erg s^{-1} (canonical estimate)
$|g_\chi| \approx |g_A^\mu|$
$m_{A'}$ [GeV]
Fraction of optically thin axial luminosity
Compton
neutrino
dimuon
$m_{A'}$ [GeV]